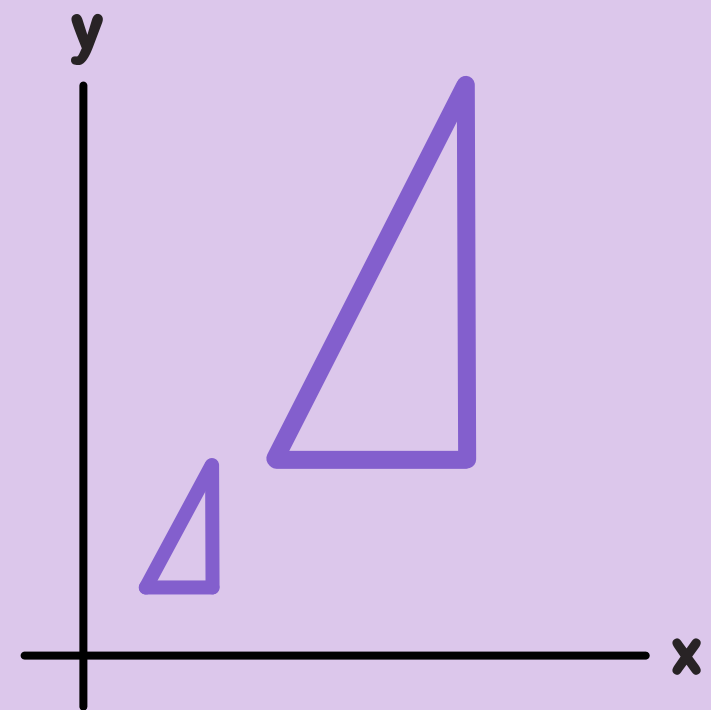
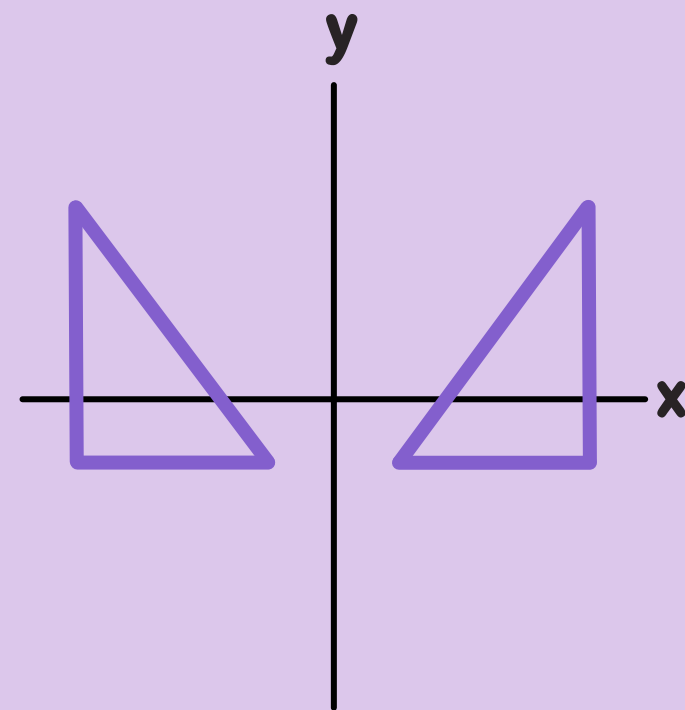
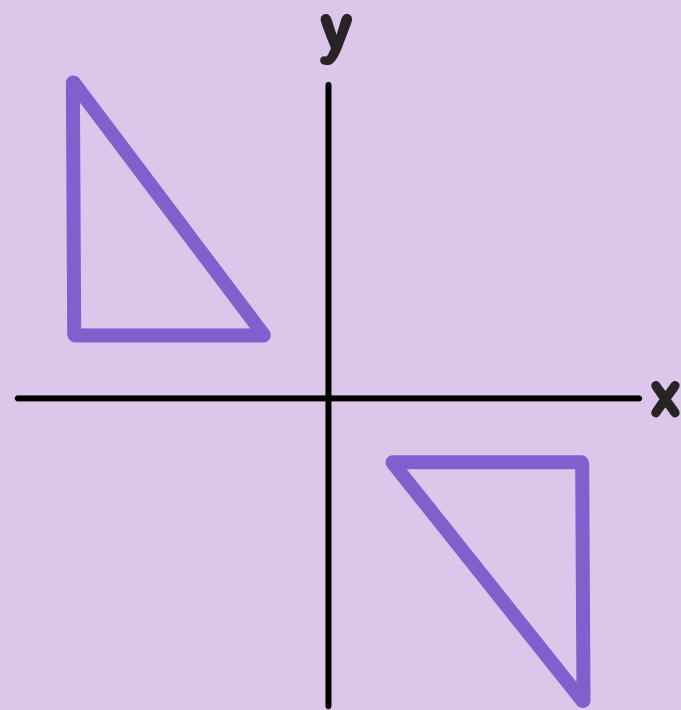
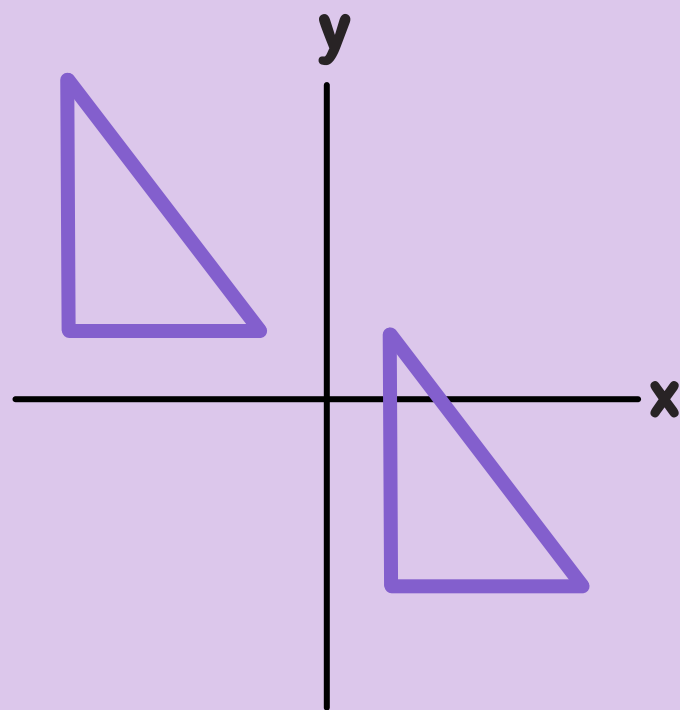


TRANSFORMATIONS

GRADE 7



**What are
Coordinates?**



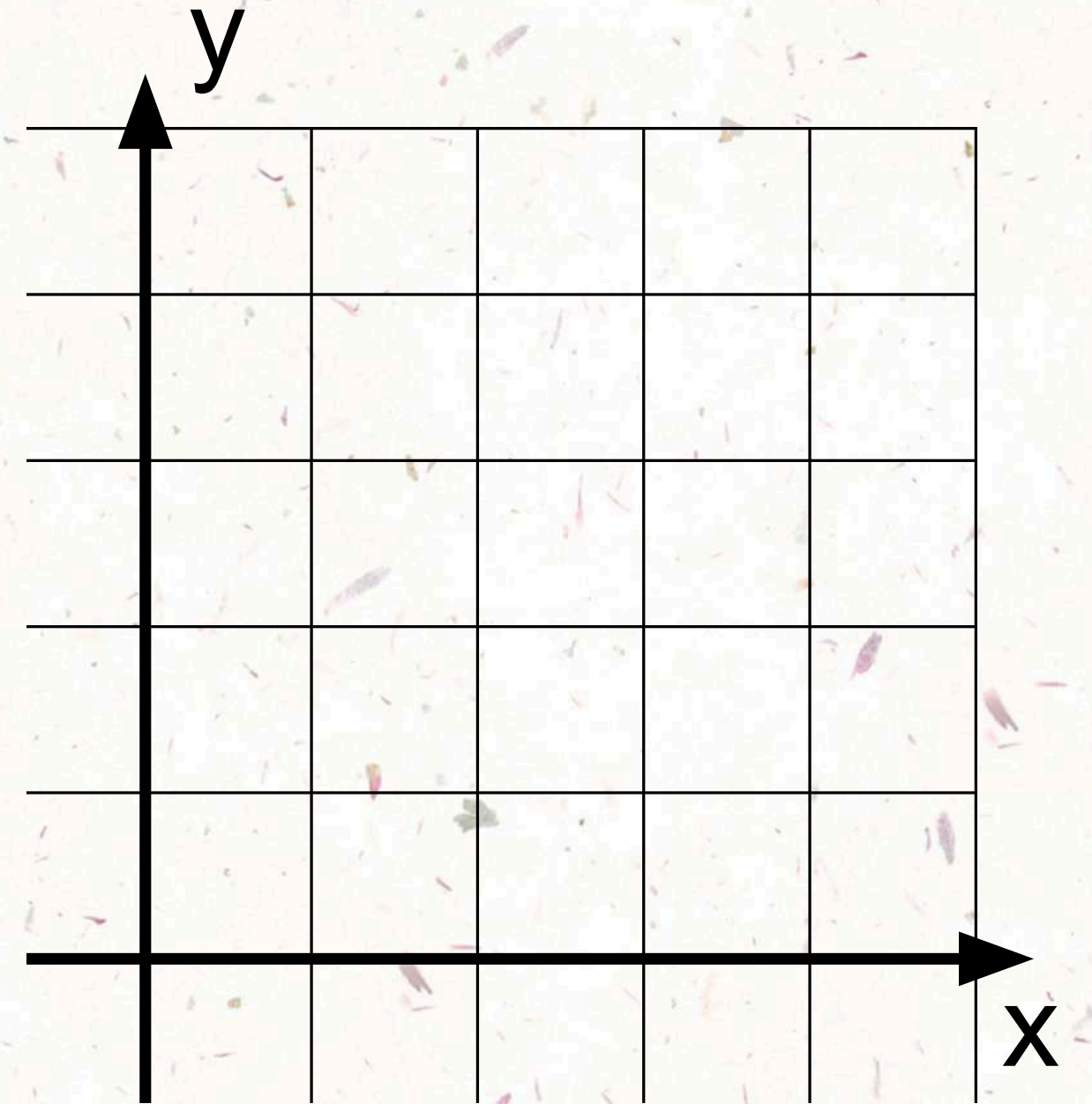
What are Coordinates?

Coordinates are pairs of numbers used to locate points on a plane. This plane is defined by two lines called axes: the x-axis (horizontal) and the y-axis (vertical), which intersect at a point called the origin.



Parts of a Coordinate

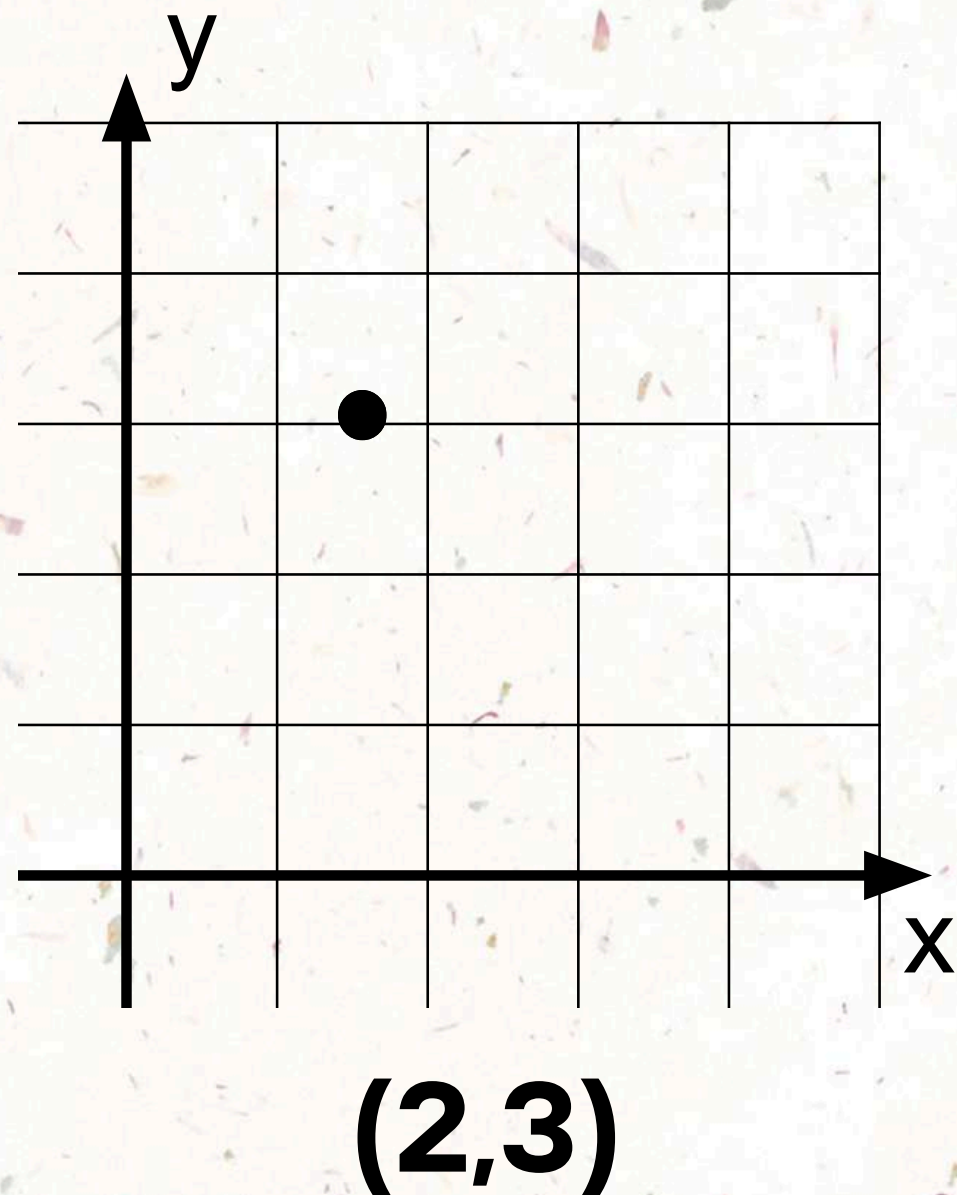
Coordinates are pairs of numbers indicating a point's horizontal (x) and vertical (y) positions from the origin, written as (x, y) .



**How do you
Make a
Coordinate?**



Plotting Coordinates



To plot a coordinate (x, y) :

Move x units horizontally and y units vertically from the origin $(0, 0)$ to locate the point (x, y) on the graph.

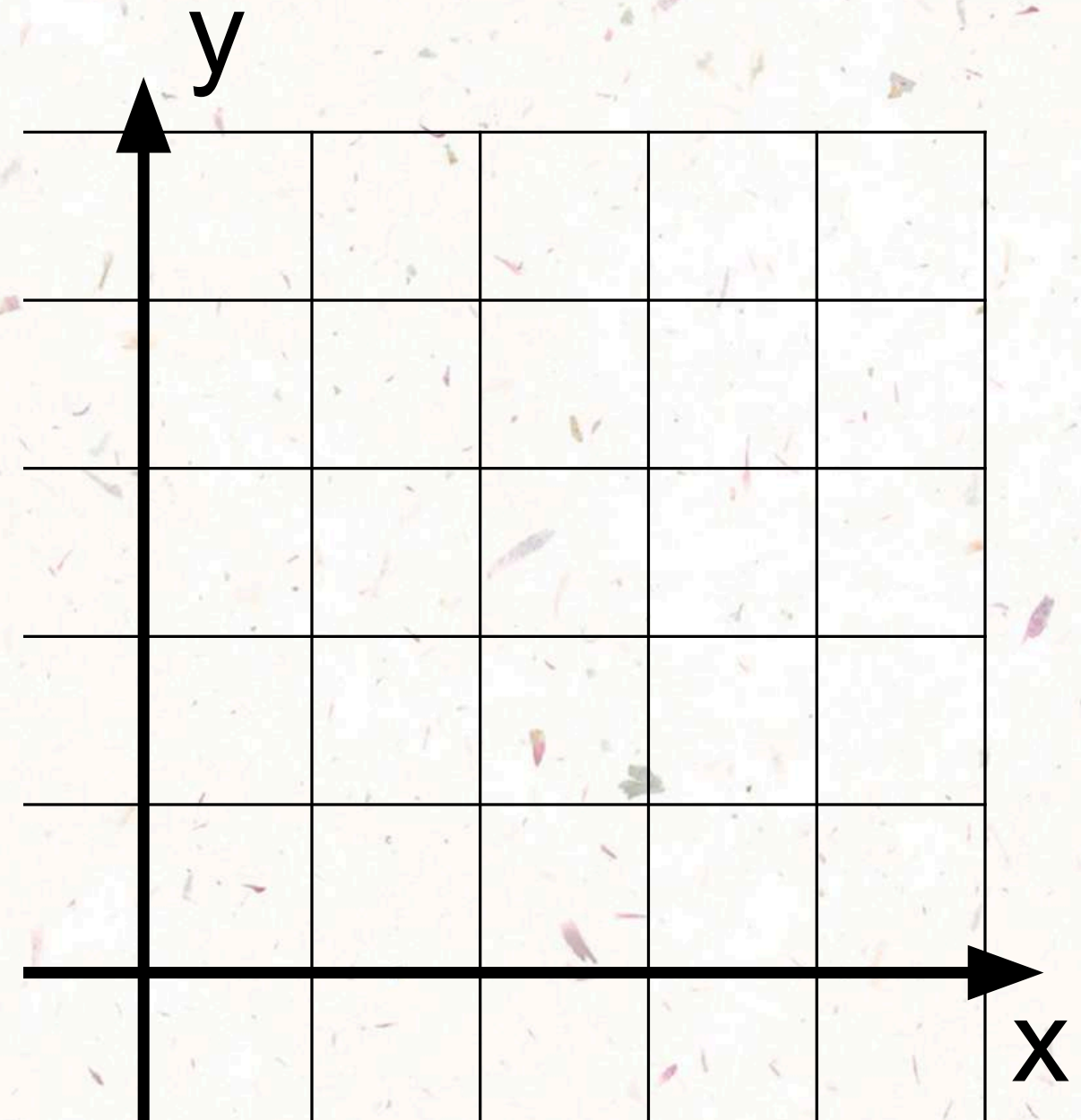


Let's Practice!

Drag the dot to the
following coordinates!



(3,4)

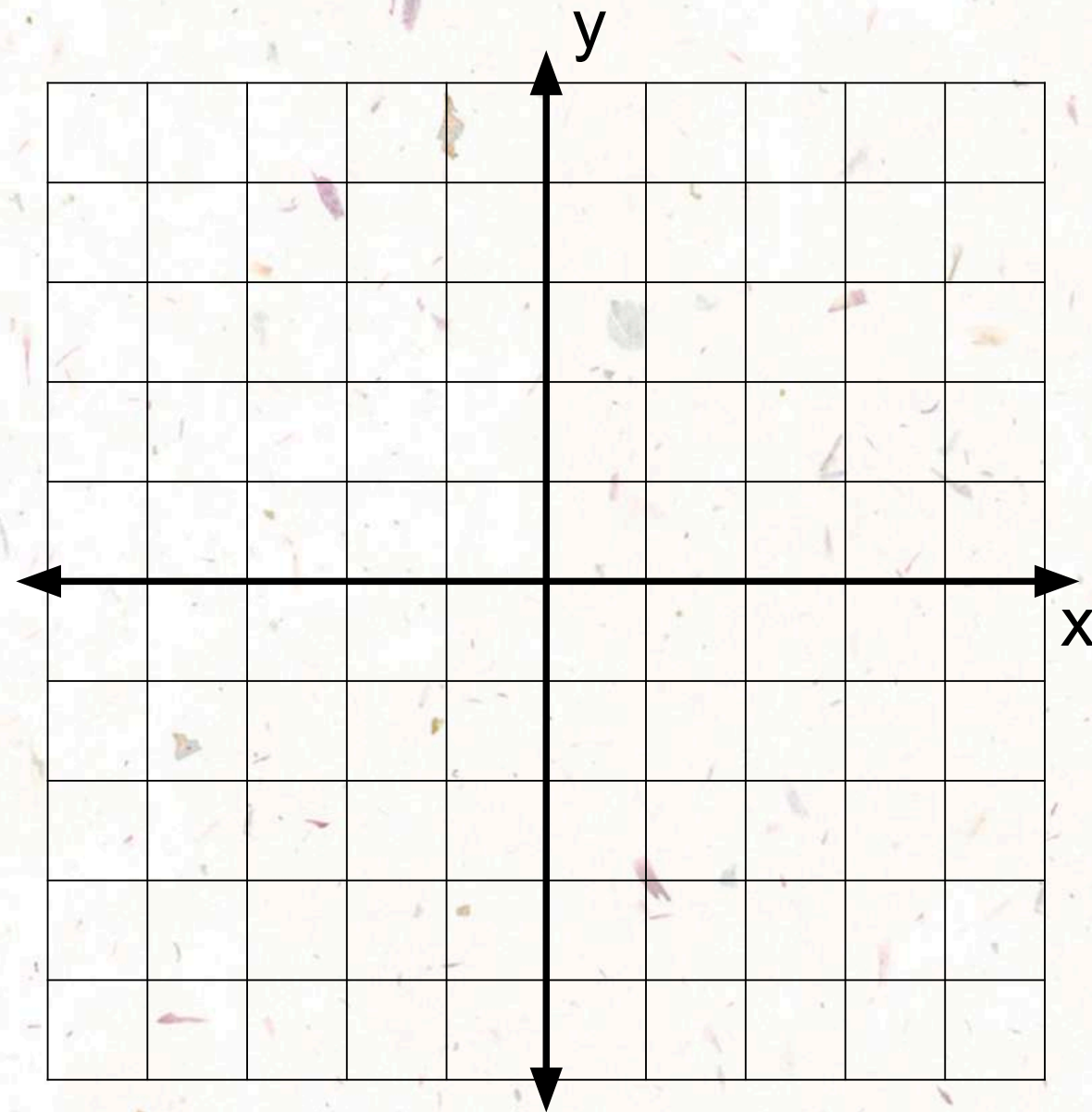


What is a Quadrant Plane?



Quadrant Plane

The coordinate plane is formed by perpendicular x-axis (horizontal) and y-axis (vertical) lines that intersect at the origin.



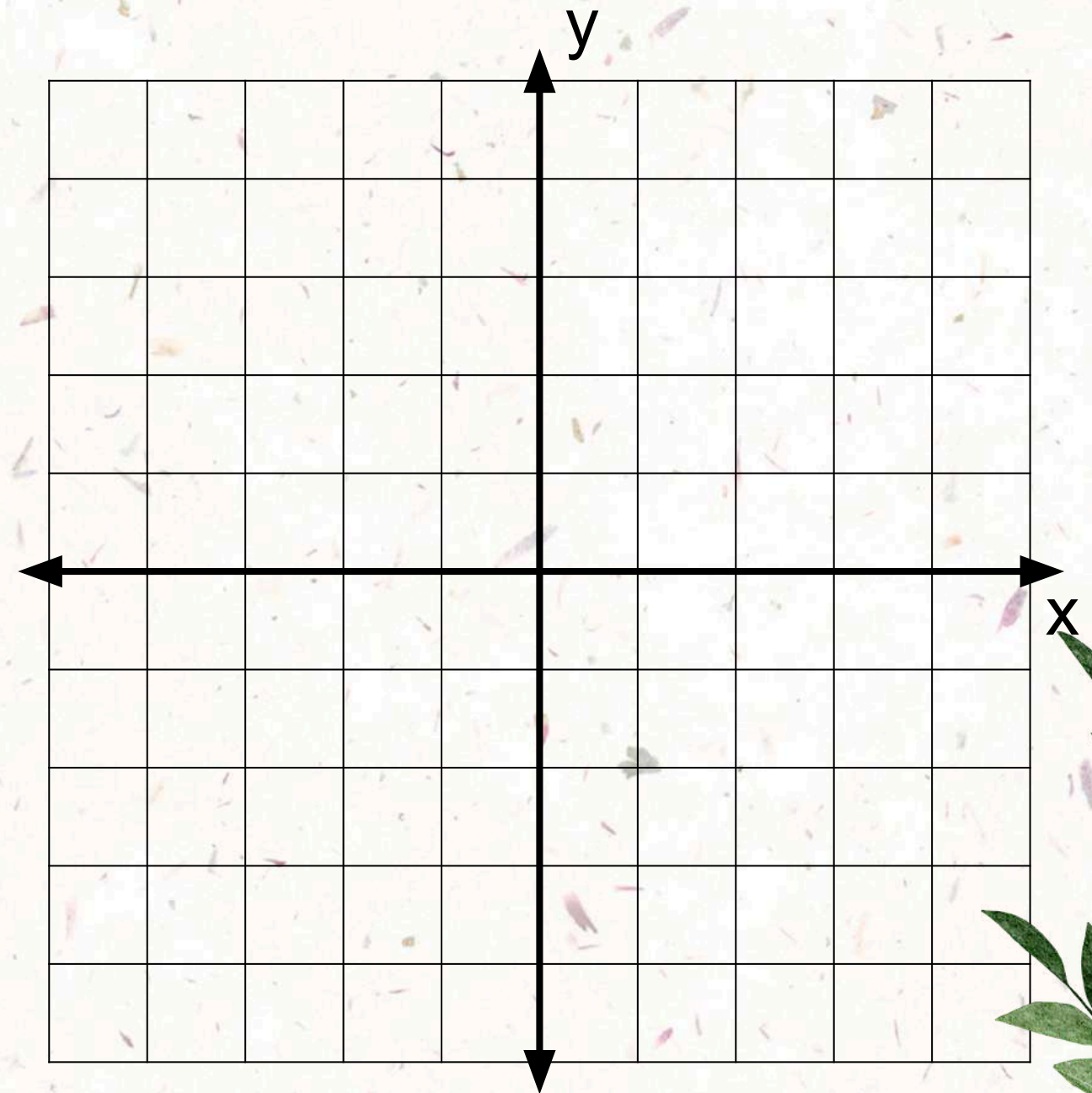
It is divided into four quadrants and is used to graphically represent pairs of numbers as points.

Let's Practice!

Drag the dot to the
following coordinates!



$(-4, 5)$

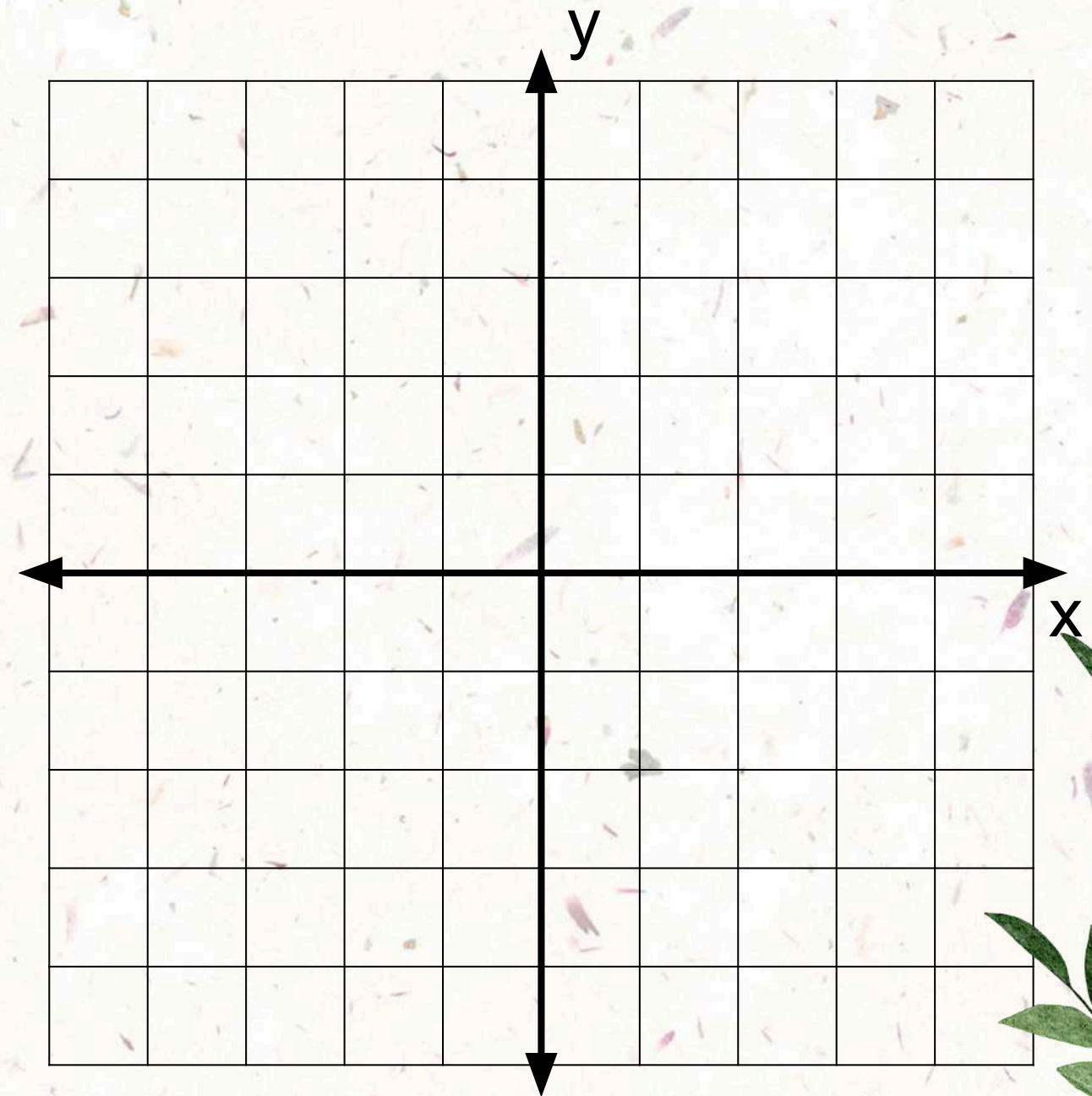


Let's Practice!

Drag the dot to the
following coordinates!



$(2, -3)$

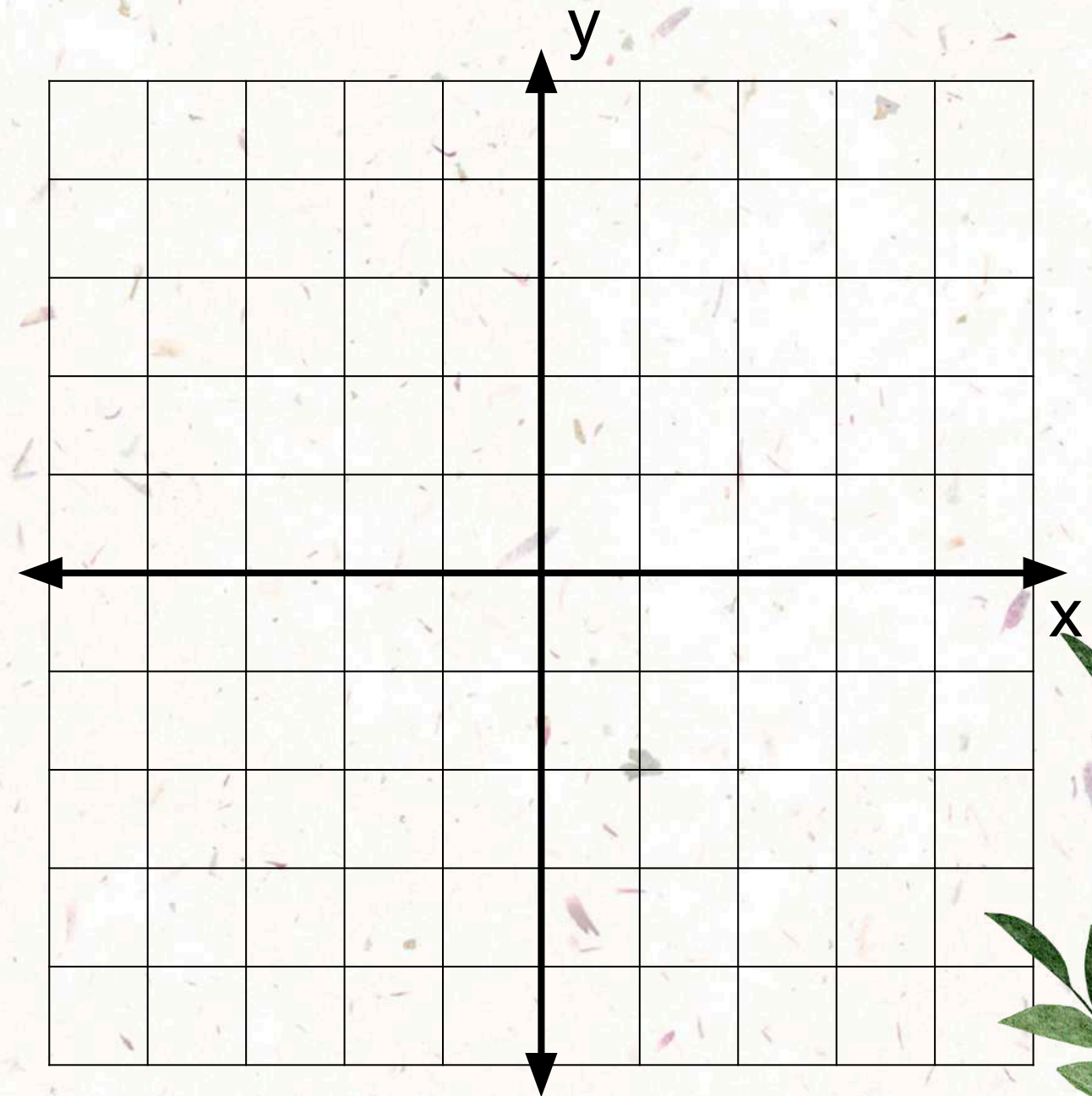


Let's Practice!

Drag the dot to the
following coordinates!



$(-4, 4)$

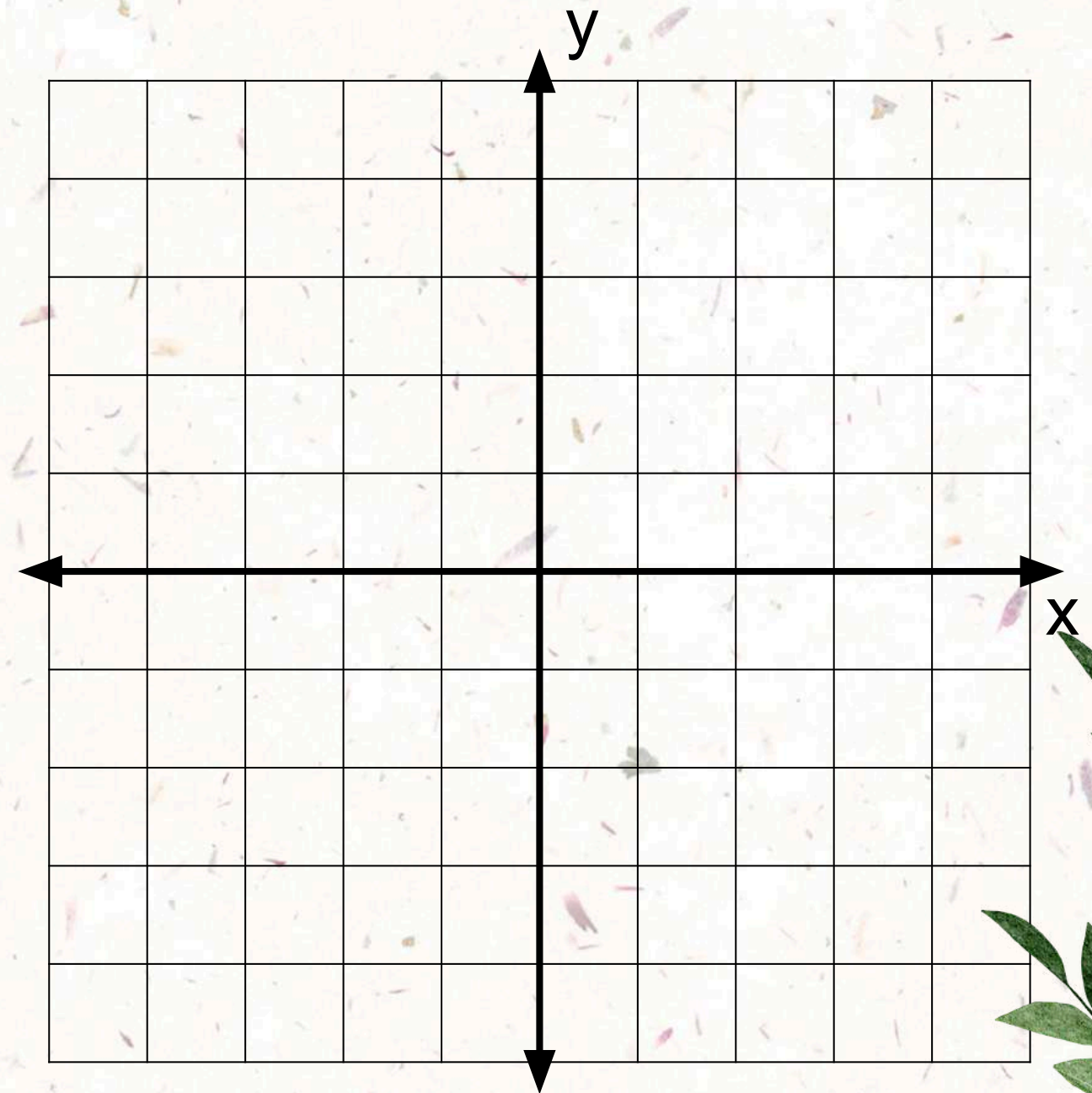


Let's Practice!

Drag the dot to the
following coordinates!



(1,-5)



QUESTION TIME



USE GRAPH PAPER TO FIND THE NEW COORDINATE OF A POINT AFTER THE FOLLOWING TRANSFORMATIONS ARE APPLIED

1.

**TRANSLATE THE POINT
(2,3) 4 UNITS RIGHT
AND 1 UNIT DOWN**

2.

**REFLECT THE POINT
(2,3) OVER THE
X-AXIS**

3.

**REFLECT THE POINT
(2,3) OVER THE
Y-AXIS**

ANSWERS



USE GRAPH PAPER TO FIND THE NEW COORDINATE OF A POINT AFTER THE FOLLOWING TRANSFORMATIONS ARE APPLIED

1.

TRANSLATE THE POINT
(2,3) 4 UNITS RIGHT
AND 1 UNIT DOWN

(6,2)

2.

REFLECT THE POINT
(2,3) OVER THE
X-AXIS

(2,-3)

3.

REFLECT THE POINT
(2,3) OVER THE
Y-AXIS

(-2,3)

PRACTICE:

USE COORDINATE RULES TO FIND THE COORDINATES OF A'B'C' AFTER A TRANSFORMATION IS APPLIED TO A TRIANGLE WITH COORDINATES A(3,-1), B(5,2), AND C(6,-3).

1. TRANSLATE ABC 7 UNITS
LEFT AND 5 UNITS UP

2. REFLECT ABC OVER THE
X-AXIS

3. REFLECT ABC OVER THE
Y-AXIS

4. ROTATE ABC 90°
CLOCKWISE ABOUT ORIGIN

5. ROTATE ABC 270°
CLOCKWISE ABOUT ORIGIN

6. DILATE ABC BY A SCALE
FACTOR OF 2

PRACTICE ANSWERS:

USE COORDINATE RULES TO FIND THE COORDINATES OF $A'B'C'$ AFTER A TRANSFORMATION IS APPLIED TO A TRIANGLE WITH COORDINATES $A(3,-1)$, $B(5,2)$, AND $C(6,-3)$.

1. TRANSLATE ABC 7 UNITS
LEFT AND 5 UNITS UP

$A'(-4,4)$ $B'(-2,7)$ $C'(-1,2)$

2. REFLECT ABC OVER THE
X-AXIS

$A'(3,1)$ $B'(5,-2)$ $C'(6,3)$

3. REFLECT ABC OVER THE
Y-AXIS

$A'(-3,-1)$ $B'(-5,2)$ $C'(-6,-3)$

4. ROTATE ABC 90°
CLOCKWISE ABOUT ORIGIN

$A'(-1,-3)$ $B'(2,-5)$ $C'(-3,-6)$

5. ROTATE ABC 270°
CLOCKWISE ABOUT ORIGIN

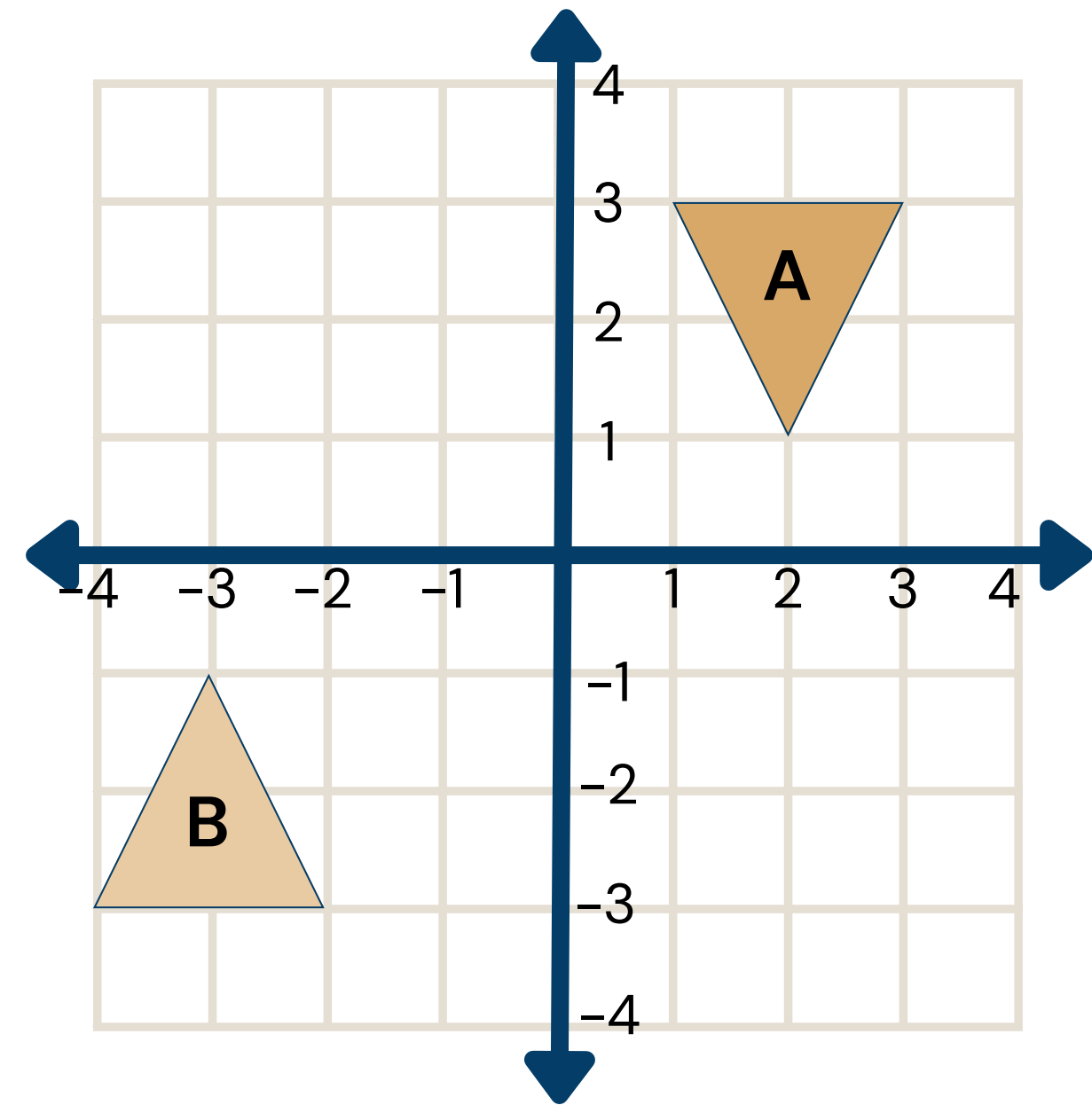
$A'(1,3)$ $B'(-2,5)$ $C'(3,6)$

6. DILATE ABC BY A SCALE
FACTOR OF 2

$A'(6,-2)$ $B'(10,4)$ $C'(12,-6)$

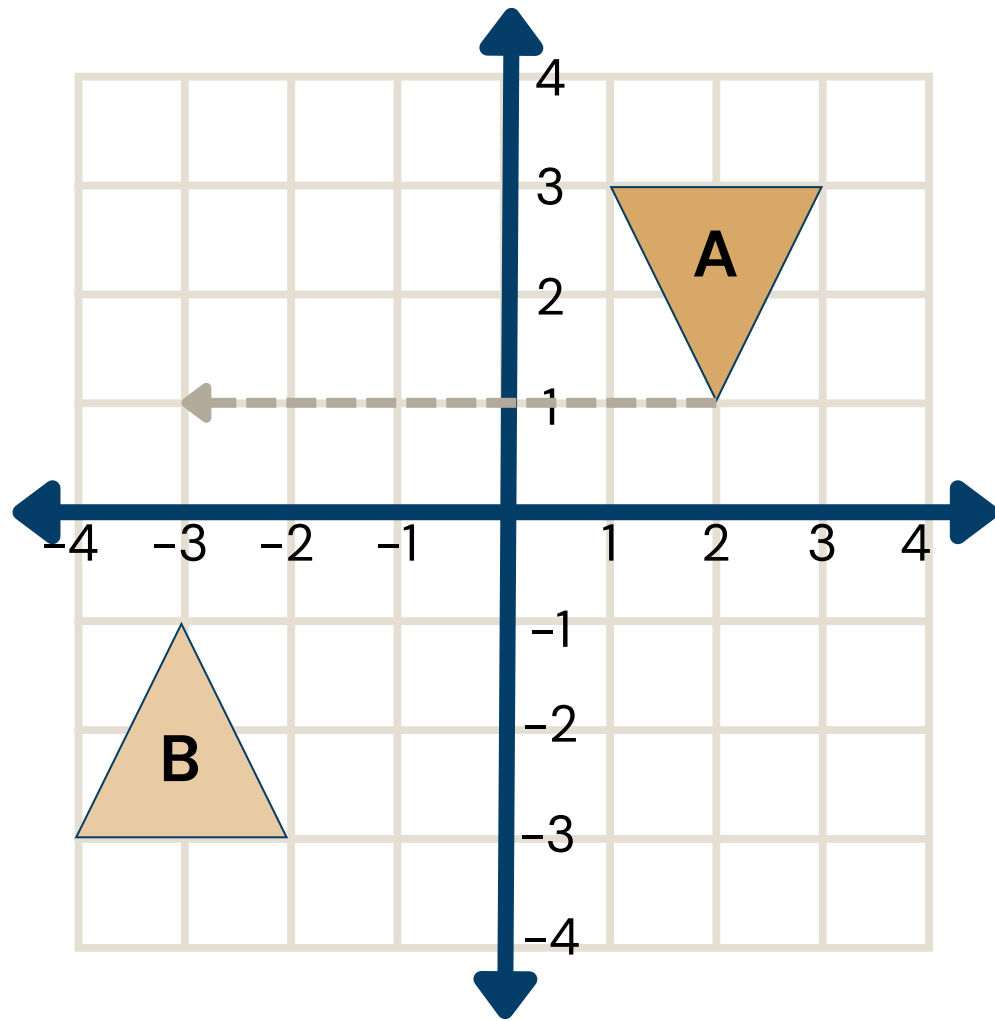
EXAMPLE: SEQUENCES OF CONGREUNCE TRANSFORMATIONS

**DESCRIBE THE SEQUENCE
THAT MAPS SHAPE A
ONTO SHAPE B**

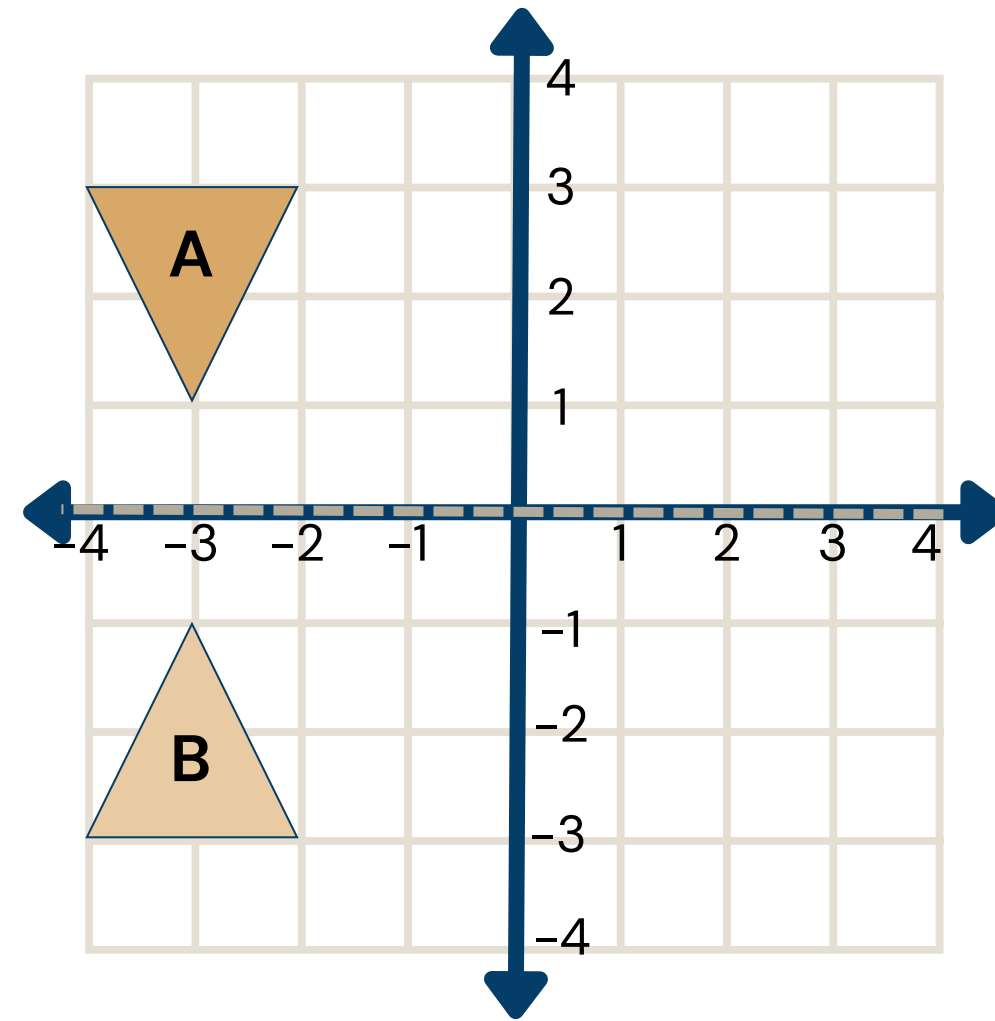


SOLUTION

STEP 1: TRANSLATION (X - 5, Y)

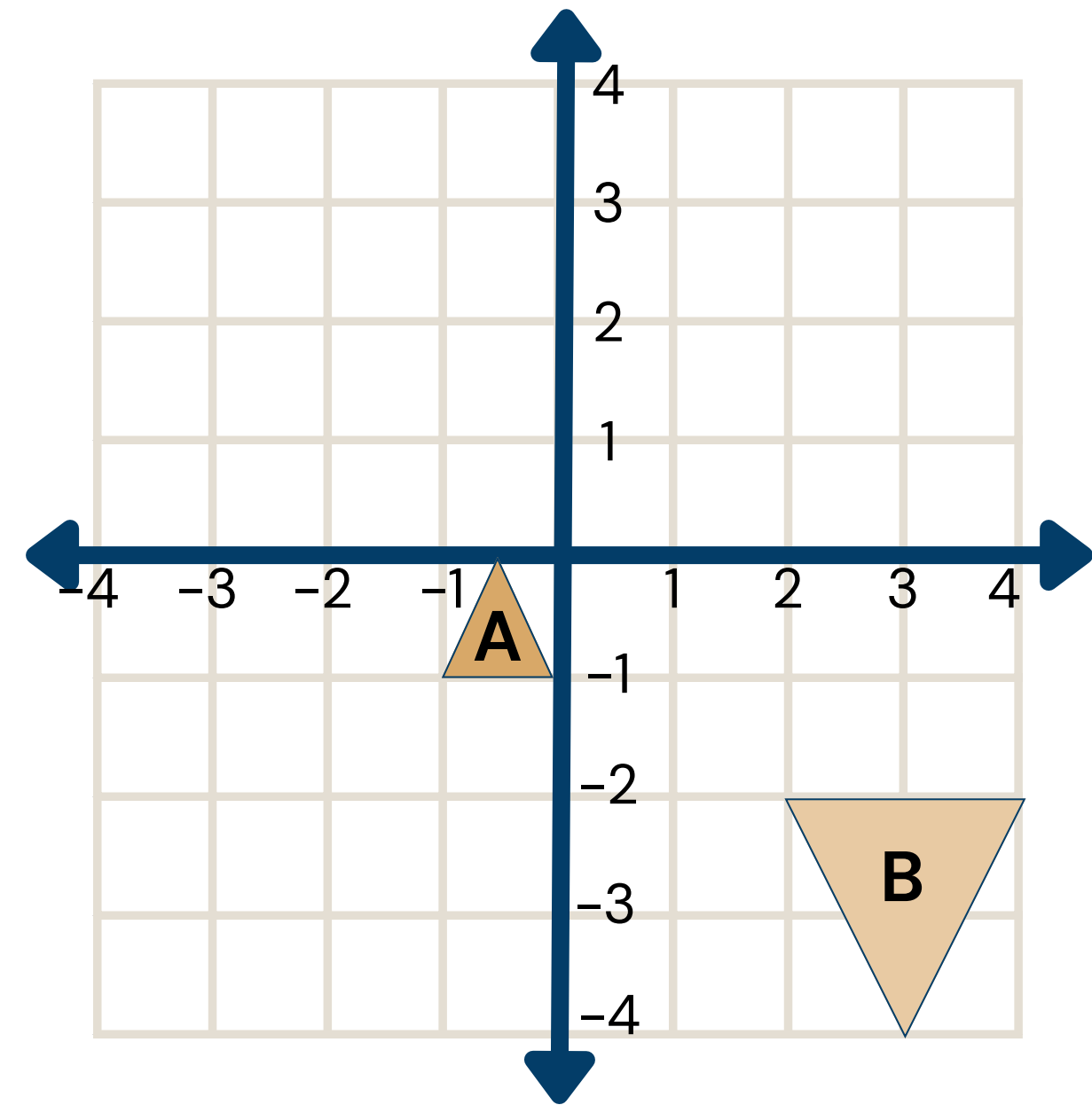


STEP 2: REFLECTION IN X AXIS



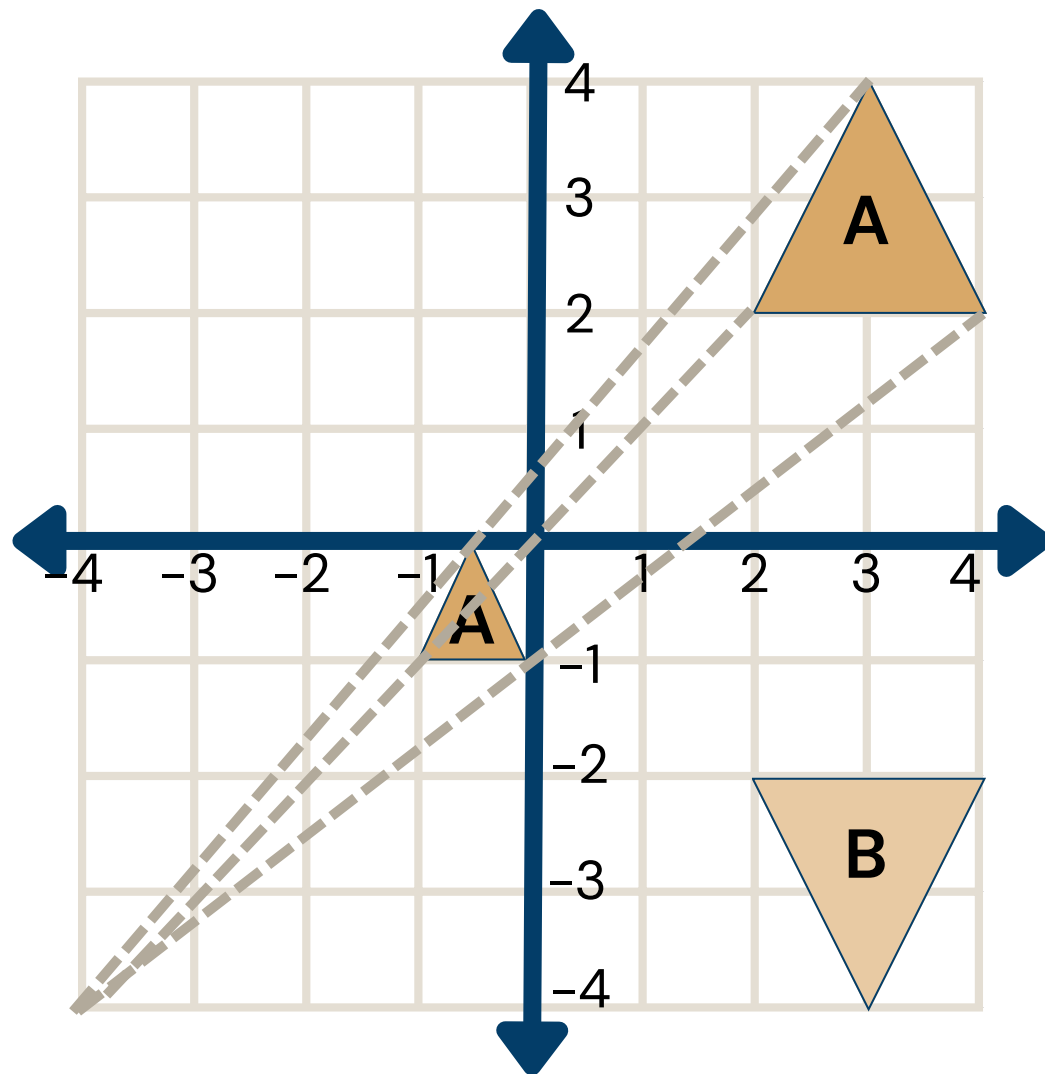
EXAMPLE: SEQUENCES OF **SIMILAR** TRANSFORMATIONS

**DESCRIBE THE SEQUENCE
THAT MAPS SHAPE A
ONTO SHAPE B**

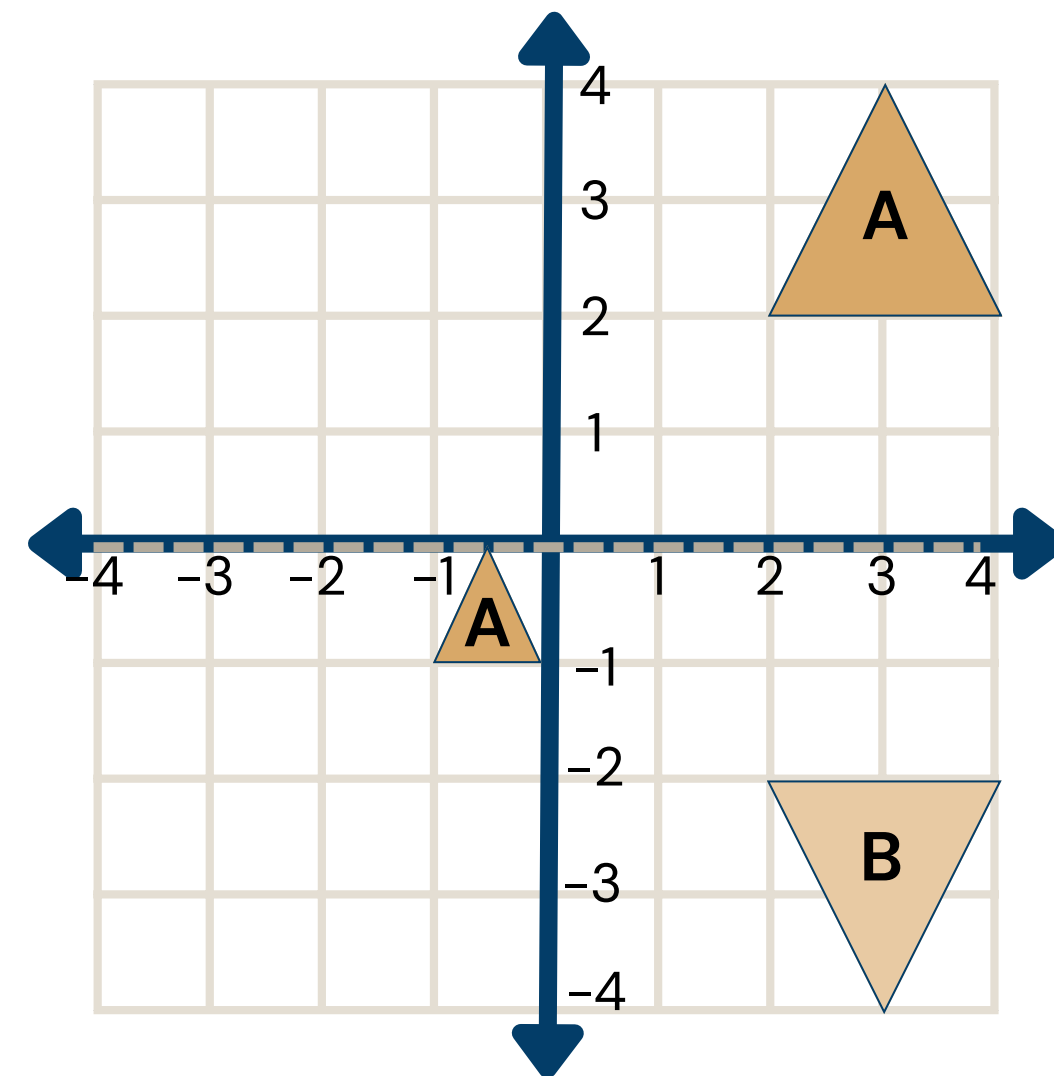


SOLUTION

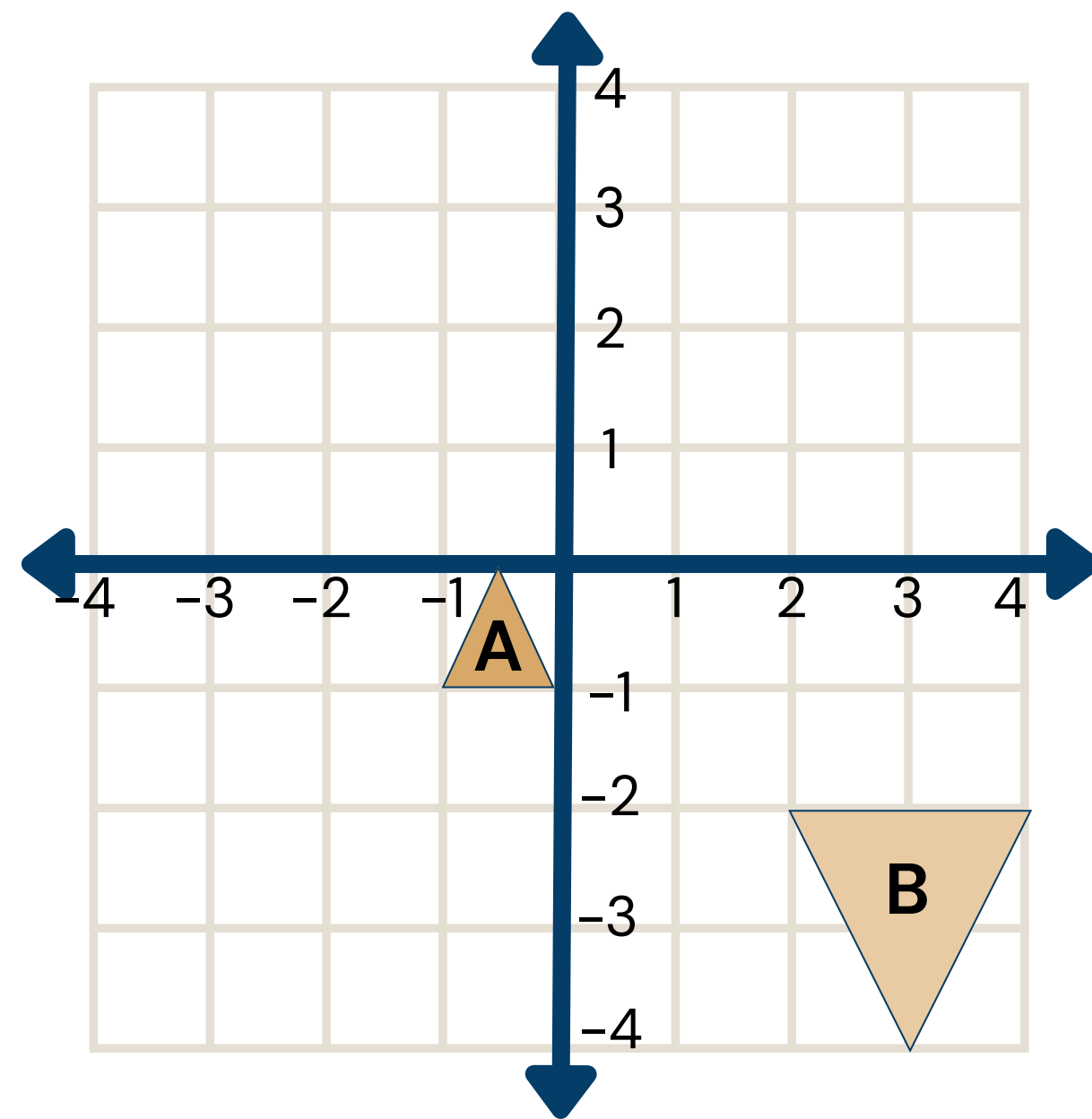
STEP 1:
DILATION S.F. = 2, CENTER $(-4, -4)$



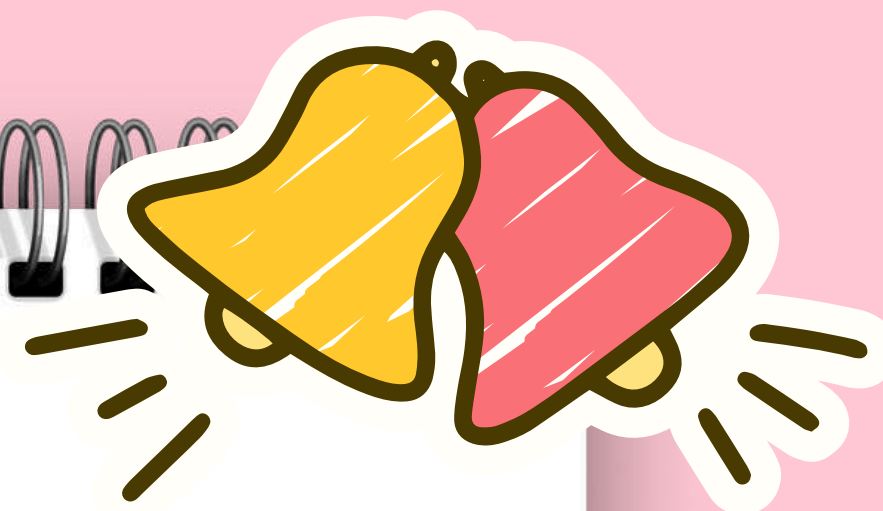
STEP 2:
REFLECTION IN X AXIS



SIMILAR SHAPES



Both shapes are similar if dilation is used as part of the sequence of transformations.



THANK YOU

Follow for more such math lessons!

